

Theme A, in plain language

ICS553 Prosit 1 · [notebooks/02_distributions.ipynb](#) · a guide to what the notebook does, what your numbers mean, and the four choices only you can make.

1. What Theme A is actually arguing

You are advising a committee on where to send a limited delivery of bed nets. Every number you give them carries uncertainty. Theme A's job is to make that uncertainty **honest**, and it does so by answering two questions.

- **Question 1 — what shape do the case counts have?** If you assume the wrong shape, the uncertainty you quote around your estimate is wrong too. Tasks A1, A2 and A3.
- **Question 2 — how wide is the interval on a survey estimate?** The household survey was not a simple random sample. Treating it as one makes your intervals look tighter than they are. Task A4.

That second point is the one that reaches your allocation rule. A district can deserve nets because its burden is high, **or** because your interval is so wide you cannot rule out a high burden. Ranking on the average alone hides that second group entirely.

2. The four tasks, one at a time

A1 — Look at the counts and notice they are badly behaved

The idea. A count is a whole number of things that happened: 41,612 confirmed malaria cases in one district. The simplest model for counts is the Poisson distribution, and it has one unusual property: its mean and its variance are **the same number**. So there is a one-line test of whether Poisson could possibly work — divide the variance by the mean. If Poisson fits, you get 1.

What the code does. `models.moment_summary()` returns the mean, the variance and their ratio, then plots a histogram.

Your result. Variance / mean = **77,183**. Poisson requires 1.0.

What it means. This is not a close call. The counts vary roughly 77,000 times more than a Poisson allows. The reason is intuitive: Poisson assumes one single rate produced every district's count, but districts differ hugely in population and in transmission intensity. When the rate itself varies, the spread grows. Statisticians call it **over-dispersion**.

If the panel asks

"The variance is 77,000 times the mean. A Poisson requires them to be equal, so it is ruled out before we fit anything. Part of that ratio is just population differences between districts, which is why we re-check it in A3 after adjusting for population."

A2 — Fit the Poisson anyway, and show it failing

The idea. You do not fit Poisson because you think it will work. You fit it so the report can show *why* you needed something better. Its one parameter, the rate, is written λ (lambda), and the best estimate of it is simply the average of your counts.

Your result. $\lambda = 195,640$. That forces a standard deviation of $\sqrt{\lambda} = 442$. Your districts actually vary with a standard deviation of **122,882** — about **278 times wider**.

What it means. The fitted Poisson draws as a near-vertical spike on the histogram. It is not that it fits badly and could be tuned; it has **no free parameter capable of fitting this**. Its spread is locked to the square root of its mean.

If the panel asks

“The Poisson's spread is not adjustable — it is fixed at the square root of the mean. Ours would be 442 where the data shows 122,882. That is why we need a distribution with a second parameter.”

A3 — The Negative Binomial, which has the missing knob

The idea. The Negative Binomial (NB) is the Poisson plus one extra parameter, called **alpha** (α), that controls how much extra spread is allowed. Its variance rule is:

$$\text{variance} = \text{mean} + \alpha \times \text{mean}^2$$

Set $\alpha = 0$ and you get the Poisson back exactly, so the two models are directly comparable. α measures precisely how much extra-Poisson variation your data demands.

Two ways the notebook gets alpha. First by algebra, rearranging that formula on your sample numbers — no optimiser, nothing to take on trust. Then by maximum likelihood, which is the number you report.

The offset, which matters more than it sounds. Comparing raw counts across districts of wildly different sizes mixes up two things: how much malaria there is, and how many people there are. The fix is to tell the model each district's population as a fixed, known quantity — an **exposure offset**. The model then describes the *rate* while the maths still treats the response as a count, which is what Poisson and NB require. It costs no parameters.

Your results.

Quantity	Value	How to read it
α by algebra (no covariates)	0.3945	Describes all spread in the raw data
α by maximum likelihood (with offset)	0.2677	Spread left <i>after</i> population is accounted for
Poisson dispersion after offset	54,147	Should be about 1. Poisson still fails.
AIC gap, Poisson → NB	2,393,669	Lower is better. One extra parameter bought this.

Why the two alphas differ — a question you will be asked. They answer different questions. The first measures *all* the variation between districts, including the fact that some are ten times larger than others. The second measures only what is left over once population is accounted for. Quote the smaller one for the model; quote the larger one when describing the raw data. Say which is which.

A trap the notebook already caught for you

net_coverage_pct is a **region-level figure** glued onto districts, so it holds only 3 distinct values — one per region. Put it in a model beside a region variable and the two are mathematically identical, the fit collapses, and statsmodels reports only a cryptic `LinAlgError`.

`models.check_design_matrix()` catches this first and names the guilty column. Your model spec came back rank 3 where it needed 5.

Consequence for your report: that coefficient is a *region* effect wearing a net-coverage label. Never report it as “districts with more nets have fewer cases”.

A4 — How wide the interval really is

The idea, and it is the most important one in Theme A. The survey did not pick households at random across Ghana. It picked **clusters** (villages or enumeration areas) first, then picked households inside them. That matters because two households in the same cluster are not independent pieces of information — they share a village, a water body, a mosquito population, and whichever net campaign last passed through.

What a bootstrap is. You want to know how much your estimate would wobble if you could repeat the survey. You cannot repeat it, so you fake it: resample your own data with replacement, thousands of times, and watch how much the answer moves. The middle 95% of those answers is your confidence interval.

The part people get wrong. If you resample *households*, you are pretending the survey drew households independently. It did not. You credit yourself with more information than the survey bought, and your interval comes out too narrow. The correct version resamples **clusters first**, then households inside the drawn clusters.

Your results (for the automatically-chosen rural domain):

Method	Interval	Width	Verdict
Naive — resample households	64.8 – 76.4%	11.6 pp	Wrong for this design
Cluster — resample clusters	58.8 – 79.7%	20.9 pp	What you report

The correct interval is **1.81 times wider**. Squaring that ratio gives the **design effect** of **3.27**, which reads as: this cluster sample carries about as much information as a simple random sample **one third its size**. If you had used the naive interval you would have overstated your precision threefold.

The validation, and its honest limit. DHS publishes its own figure for regional net ownership. Your survey-weighted estimate reproduced it to **0.03 percentage points**. That is a genuine result: it validates your weighting and your estimator. It does **not** validate your interval, because the published intervals are missing from the file (see section 5). Claim the first, not the second.

If the panel asks

“Resampling households assumes the survey drew them independently. It drew clusters first, so households inside one cluster are correlated. Our cluster bootstrap gives an interval 1.8 times wider, a design effect of 3.3. We also reproduce the DHS published point estimate to 0.03 percentage points, which validates our weighting.”

3. The four choices only you can make

The notebook deliberately stops short in four places. These are the parts a panel will probe, and borrowed answers will not survive the follow-up question.

Choice 1 — Override the A4 domain (do this first; it changes your numbers)

The notebook picks the region with the fewest rural clusters automatically. On your data that resolved to **rural Greater Accra** — 9 clusters, 257 households.

Problem: Greater Accra is metropolitan. It is technically under-sampled for rural households, but it is not the equity story, and it is disconnected from your northern district case data.

Recommendation: set `target_region` by hand to Northern (12), Upper East (15) or Upper West (16), so the survey domain and the case data describe the same place. Write one sentence in the notebook saying you overrode it and why.

Then re-run — the design effect, both intervals and the claims table all change.

Choice 2 — Describe the shape of the distribution (A1)

Look at your histogram and say in one sentence: where is most of the mass, how long is the right tail, and is any single district far enough out that you would want to check it against the original workbook before trusting it?

Why it is asked: an outlier in routine surveillance data is often a reporting artefact, not a real epidemic. Noticing that is the analyst's job.

Choice 3 — Compare your interval to the published one (A4.1)

State whether the intervals overlap, and whether yours is wider or narrower.

Either answer is fine, but you must say why. Two reasons yours may be wider: your bootstrap also resamples households within clusters, adding variance; and DHS uses the full survey design, including stratification information you do not have.

Choice 4 — State your claim at the strength the evidence supports (A4.1)

“Our weighted estimator reproduces the DHS published figure to within 0.03 pp” — **defensible**.

“Our confidence interval is validated” — **not defensible**. There is no published interval in the file to compare against.

Knowing the difference is most of what the rigour mark is for.

4. The jargon, in one line each

Term	What it means
Count data	Whole numbers of events. Cannot be negative or fractional.
Over-dispersion	The data varies more than the model allows. Here: variance far above mean.
λ (lambda)	The Poisson's only parameter, its rate. Estimated by the sample mean.
α (alpha)	The NB's extra parameter. How much spread beyond Poisson. $\alpha=0$ is Poisson.
Offset / exposure	Population fed to the model as a known quantity, so it models a rate not a raw count.
Dispersion ratio	Pearson chi-squared divided by degrees of freedom. About 1 means the model fits.
AIC	Fit score that penalises extra parameters. Lower is better; only compare on the same data.
Bootstrap	Resample your data thousands of times to see how much the answer wobbles.
Cluster	The village or enumeration area the survey sampled as a unit. Not a district.
Design effect (DEFF)	How much precision the clustering costs you. 3.3 means you effectively have a third of the sample.
pp	Percentage points. The gap between 60% and 65% is 5 pp, not 5%.

5. Data problems you must mention in the report

These came out of running the pipeline. Each is worth marks — a documented discrepancy is a finding, not a failure.

#	What is wrong	What it costs you
1	The survey file has no parasitaemia column and no district identifier , though the case brief describes both.	A4 estimates net ownership for a <i>region</i> , not prevalence for a district. Say “region” in the report.
2	The published ITN confidence intervals are empty — 0 values in 76 rows. Prevalence CIs are populated.	The promised check against the DHS reference cannot close. You validate the point estimate only.
3	net_coverage_pct is region-level (3 values for 3 regions) and measured in 2022 against 2014–17 cases.	Cannot sit in a model beside a region variable. Its coefficient is not a district net effect.
4	months_reported, year_start and year_end are constant.	No information in them. B1's data-gap signal must come from somewhere else.
5	ghana_region_crosswalk.csv is missing from the course package.	Needed in B1 to join survey regions to map polygons. It is public; source it separately.
6	7 of 50 district names do not match the boundary files — mostly districts split after 2018.	B1 needs a hand-built name-to-code crosswalk. The 7 unmatched are themselves a reportable gap.

6. Running it

```
conda activate prosit1
jupyter lab # kernel: Python 3.11 (prosit1)

# headless check that everything still runs, without touching the repo copy
jupyter nbconvert --to notebook --execute \
--output-dir /tmp/prosit-run --output 02_smoke.ipynb \
notebooks/02_distributions.ipynb

# always strip outputs before committing
jupyter nbconvert --clear-output --inplace notebooks/02_distributions.ipynb
black src/ && ruff check src/
```

Where the code lives. `src/models.py` holds the Poisson and NB fitting, `src/uncertainty.py` the two bootstraps, `src/io.py` the loaders, the one stratified split and the column contracts, `src/viz.py` the plots. The notebook imports and calls — it does not define analysis logic.

One rule that overrides convenience. Never paste survey rows into any AI tool, not even a single `.head()` while debugging. Paste the error message, the column names, or aggregate output instead. This is in the case brief and in `RULES.md` section 1.